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RESEARCH

Buffer Zones, Population, Ecology, and Conflict: Ethnological Modeling and Archaeological Implications

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Abstract
Buffer zones, areas unoccupied due to conflict between social collectives, have been theorized to indicate the existence of population pressure as a motivator for conflict in small-scale subsistence economies. Using a database of ethnographic and historical buffer zones, we show that buffer zone width is primarily influenced by population density and not by ecology or subsistence mode, suggesting that security concerns primarily account for buffer zone dynamics. Buffer zones may still contribute to resource imbalances, however. Buffer zones appear to be most effective in the context of intercommunity raiding but may also be used more deliberately by states to monitor the movement of people or defend against internal or other non-state actors. Archaeologists should therefore expect them to exist under certain conditions of ancient conflict. However, identifying buffer zones in the archaeological record is challenging. We review a variety of approaches to examining ancient buffer zones, including using population estimates to model the plausible size of ancient buffer zones.

Keywords Buffer zones · Collective violence · Warfare · Ecology · Population density

Introduction

Buffer zones, areas that separate combatants with the goal of increasing security and/or reducing the likelihood of violence, have received comparatively little archaeological attention as a signature of collective violence. Anthropologically, interest in buffer zones developed within the broader theoretical framework of cultural ecology and was spurred by Hickerson's (1965, 1970) ethnohistoric work on conflict between the Dakota and Chippewa in the Upper Great Lakes region between the seventeenth and

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nineteenth centuries. He argued that buffer zones between the Dakota and Chippewa allowed for regeneration of Virginia Deer, a critical resource, and prevented over-hunting and resultant subsistence stress. Hickerson's model was subsequently applied to understanding population-resource balance in the Amazon as part of a vehement debate over the role of protein scarcity in stimulating warfare, particularly among the Yąnomamö of the Brazil-Venezuela border region (Chagnon & Hames, 1979, 1980; Harris, 1977; Lizot, 1977; Ross, 1980; Ross & Ross, 1980). DeBoer (1981) estimated that up to half of potentially settled and utilized area around rivers in the Amazon Basin may have remained unoccupied or under-utilized due to hostilities, and that defense of these areas indicated the degree to which resources there were essential for maintaining population levels. In the context of cultural-ecological models of warfare, conflict and its outcomes were viewed as an adaptive response to living in resource-limited environments.

Cultural ecological adaptive models of warfare largely lost anthropological favor as theoretical paradigms shifted after the 1970s. However, as the pace of anthropogenic climate change and environmental impact on modern conflicts have become ever more apparent, the dynamics of population and resource availability has reemerged as a significant topic both in broader studies of conflict (Adams *et al.*, 2023; Hsiang *et al.*, 2013; Turchin & Korotayev, 2005; White *et al.*, 2025) and specifically in the study of ancient violence (Gronenborn, 2007; Kennett *et al.*, 2022; Kohler *et al.,* 2009; McCool *et al.*, 2022; VanDerwarker & Wilson, 2016). This includes continued discussion of the role that buffer zones might play in the connected dynamics of population, resource availability, and conflict. For instance, Martin and Szuter (1999) have applied Hickerson's model to explain areas of high game density noted during the Lewis and Clark expedition across western North America. Bayham and colleagues (2019) have recently proposed a formal model of territorial boundary behavior among hunter-gatherers based in large part on Hickerson's buffer model. LeBlanc (2007) suggests that buffer zones were a ubiquitous feature of boundary formation (see also Myers, 1976 and Peterson, 1976) tied to the near universality of conflict in small-scale subsistence economies (LeBlanc & Register, 2003) and played a role in maintaining population below theoretical carrying capacity (Read & LeBlanc, 2003). Kondor *et al.* (2023, 2024) have recently modeled buffer zone formation (what they term "landscapes of fear") in the context of early Neolithic population dynamics, suggesting that the ceding of land and resulting loss of productivity, combined with population attrition from violence, explains modeled "boom-and-bust" population dynamics during the Neolithic period in Europe (Shennan & Edinborough, 2007; Shennan *et al.*, 2013). If these reconstructions of buffer zone dynamics are supportable, the identification of ancient buffer zones from the archaeological contexts (if possible) could have the potential to inform not just on the existence of conflict in the past, but also on the causal dynamics of ancient violence.

It is certainly the case that settlement buffer zones can produce ecological regeneration, an outcome noted in modern buffer zones like the Korean DMZ or Cyrus "Green Line" (Coates, 2014; Healy, 2007). Similar buffer zones have also been noted among other animal species, for instance, wolves, where lack of hunting along territorial boundaries can lead to higher game density (Mech, 1977, 1994). That

said, there have been numerous critiques of the "Hickerson" model of buffer zone formation, including whether or not population pressure actually existed in instances where buffer zones were observed (Chagnon & Hames, 1979, 1980), whether other historical, social, and economic dynamics better explain conflict than resource scarcity (Hamilton, 1985), and whether observed buffer zones were areas that already had higher ecological potential and thus were sources of violence rather than outcomes (Laliberte & Ripple, 2003; Lyman & Wolverton, 2002). Additionally, some scholars of warfare suggest that buffer zones are simply expedient responses to immediate threat, and that any ecological impacts of buffer zone formation are incidental outcomes of security concerns (Keeley, 1996; Kelly, 2000). The loss of valuable hunting or farming land could consequently result in either lower population than otherwise theoretically possible, or could produce population-resource stress where it was not formerly present (Kelly, 2000).

We have two goals in this paper. The first is to examine the relationship between demographic, ecological, and social variables that might influence the formation of buffer zones. We use a database of ethnographically and historically documented buffer zones to test the relative influence of population, ecological potential, subsistence mode, relative militarization, and speed of transport on buffer zone width to partially test aspects of different proposed models of buffer zone formation. We find that buffer formation is primarily an outcome of security concerns rather than the ecological potential of ceded territory, and is tightly constrained by population density. We then examine the potential to identify buffer zones in the archaeological record by examining proposed prehistoric examples and how closely they conform to expected dynamics of buffer zone formation derived from our analysis.

Buffer Zones

The term "buffer zone" has been used in a variety of contexts. As defined by DeBoer (1981: 365), "a buffer zone is simply an uninhabited area which separates two or more competing human groups." This definition is non-specific as to the nature of competition, and in its most general sense, a buffer zone can simply be a region in which two social collectives have competing interests in resource acquisition and use but do not actively settle, without necessarily implying the presence of violence. In modern geopolitics, buffer zones may refer to several related but distinct concepts of territorial usage. For instance, buffer zones may be areas in which the military of one or both sides of a conflict or potential conflict are prohibited, sometimes also referred to as exclusion zones or *cordon sanitaire*, and may be created either bi- or unilaterally (Beehner & Meibauer, 2016). Nation states or empires may also establish or support so-called buffer states, which are intended to provide border security while bearing the brunt of actual or potential violence or to prevent direct hostilities between major powers. Such states exist in modern geopolitics but were also present in the past, for instance, the "Spanish March" created in eastern Iberia by the Carolingian Empire in the eighth and ninth centuries AD (Chandler, 2013).

What we henceforth refer to as "settlement" buffer zones (also sometimes referred to as "no-man's lands" or "war zones" in the literature) are areas that could be settled

(and often were once settled), but which become under-settled and utilized or un-settled and utilized due to active or potential violence between people living on either side of the buffer. Other areas may serve as natural buffers for ecological or topographic reasons (rivers, swamps, mountains), but are always unoccupied or relatively sparsely settled and utilized. Settlement buffer zones may, however, include such features. For instance, as early as 1000 AD, the Yalu River was recognized as the formal boundary between the Chinese and Korean states. Because of ongoing tensions and potential for violence, an 80-km-wide settlement buffer zone was established around the Yalu, backed with a palisade (McKune, 1946). Regardless of type, buffer zones follow a logic of providing protection from violence and limiting its potential occurrence by putting physical distance between combatants or potential combatants (*e.g.*, Roscoe, 2011), although some settlement buffers are created to restrict movement of people across political borders in the absence of direct conflict (*e.g.*, the Melilla and Ceuta border fences, or the Inner-German border). All such buffer zones have in common a tendency to be patrolled, guarded, or otherwise monitored on both sides, whether this be by roving war parties, informal militias, state police forces or military units, fortified walls, guard towers, or with modern technological methods of surveillance.

Testing Buffer Zone Dynamics

Our analytical goal is to test certain aspects of proposed relationships between buffer zones, population, ecology, and other variables potentially relevant to buffer zones, particularly focusing on buffer width as a measure of how much territory is ceded. Given the multiple ways in which population has been invoked in relationship to buffer zones, we might anticipate some relationship between population density and buffer zone size—Keeley (1996), for instance, suggests that buffer zones should be narrower when population density is higher owing to the ability to more effectively observe and police the buffer zone, and the ability to more rapidly amass fighters to defend against any incursions across the buffer. The population predictions of "ecological" models of buffer zone formation are more difficult to test as they generally refer to either population stability or else to population that is lower than what would be theoretically possible in the absence of a buffer zone, values that are difficult at best to model or measure. As a partial test, we instead examine whether the relative value of ceded land, combined with the mode of extracting resources, relates to how much land is ceded. We additionally include variables that independently assess the relative size of fighting groups after controlling for population size, as well as a measure of travel speed. We recognize that these variables may not be exhaustive of all relevant factors that influence how buffer zones form, but they are variables that are relevant to testing existing models in the literature.

Our analysis relies on an expanded database that builds on seven cases presented by Keeley (1996: Table 7.2). Our expanded dataset (see Fig. 1) includes 52 cases of ethnographically and historically documented buffer zones for which we could measure both width (in km) and local population density (as persons per km^2) on one or both sides of the buffer zone. We also have a number of cases for which we could not estimate width and/or population density, which are included in our database but not analysis, as well as data on military exclusion zones, which we anticipate should

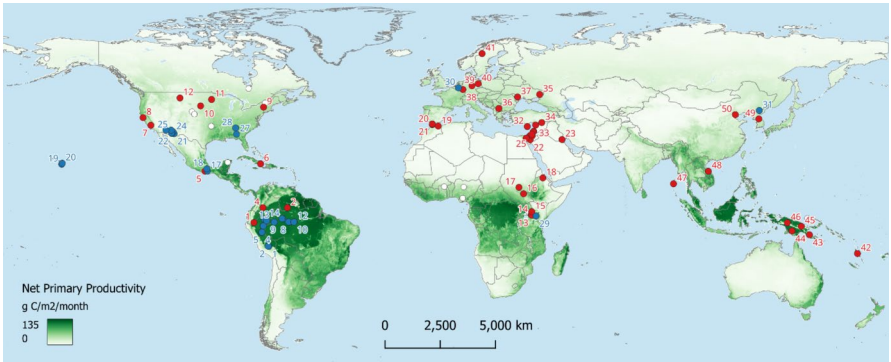

Fig. 1 Global map showing the location of buffer zones included in the present analysis. Ethnographically and historically documented buffers included in the primary analysis are indicated in red, while historical and archaeological buffers are shown in blue (see ID numbers in online supplemental data). White circles indicate buffer zones for which we have insufficient data to include in any analysis. Terrestrial net primary productivity (NPP, grams of carbon per square meter per month) is indicated in shades of green (Imhoff & Bounoua, 2006; Imhoff *et al.*, 2004). Robinson projection

show different dynamics of formation from settlement buffers. Additionally, we have collected published examples of proposed ancient (archaeological or ethnohistorically documented) buffer zones to examine existing approaches to identifying their presence. Ethnographic cases were located primarily (although not exclusively) from a search of the eHRAF database (https://ehrafworldcultures.yale.edu/), while ancient, historical, modern cases were taken from primary literature or other online sources.

Buffer Zone Width

Buffer zones were georeferenced in QGIS to the extent possible, noting that the extent of these buffer zones is often approximate. Maximal, minimal, and average widths were extracted based on the resulting polygons. As noted, these buffers are divided into "settlement" buffers, from which population is relocated or excluded, and "military" buffers, from which only active combatants and military equipment are excluded. The locations of all identified buffer zones are shown on Fig. 1, and all data are provided in supplemental files.

Population Density

Population densities were estimated for both sides of buffer zones when possible, and averaged for our analysis. For historical data, we were able to use census data to examine population densities in the immediate proximity of buffer zones, rather than averaging across entire modern nation states, while for ethnographic cases, estimates are based either on the values provided by ethnographers or by government censuses from the period in which ethnographic observation dates divided by the total area of

the territory described by those sources. We recognize that these density estimates in some cases may be approximate.

Ecological Productivity of Buffer Zone Area

As a measure of ecological potential of buffer zones, we utilize grid square estimates of net primary productivity (NPP) published and made available as GIS raster files by Imhoff and colleagues (Imhoff & Bounoua, 2006; Imhoff *et al.*, 2004). These values are derived from satellite data and reported as total carbon per 0.25 degree grid square. We have converted these into the more conventional means of reporting primary productivity in gm C/m^2/month, and utilize median values for each buffer zone.

Subsistence Mode

NPP represents only a theoretical productivity of the landscape, not what humans are actually able to extract. Humans are estimated to use somewhere between ~ 3 to 35% of all global NPP (Imhoff & Bounoua, 2006), however, this is highly variable depending on subsistence practices and probably changed considerably over the course of the twentieth century as more intensive forms of agriculture and globalized transport of food developed. As such, available data on human appropriation of NPP (HANPP) is unlikely to adequately represent how people living in subsistence economies were able to extract resources. Both measures, of course, also do not provide an assessment of the abundance of particularly critical resources. As an approximation of the relative value of a particular parcel of land, we include a subsistence variable in our analyses, based on the coding employed in the eHRAF database. Our data are coded as follows: Hunting-gathering = 1, Pastoralism = 2, Agropastoralism = 3, Horticulture/Agriculture = 4, Industrial agriculture = 5. In cases where combatants practice different subsistence modes, their respective values are averaged.

War Group Size and Militarization

We test for the effect of war group/military size using estimates of total war group size in particular conflicts. However, as other studies (Falk & Hildebolt, 2017; Oka *et al.*, 2017) have shown a clear relationship between population size and war group size, we also examine the relationship between buffer zone width and demographic conflict investment (DCI; Oka *et al.*, 2017), a measure that controls for the impact of population size scaling on war group size to model how relatively large war groups are for a given population level. We might anticipate that relatively larger war group sizes should decrease buffer width given the ability to more effectively monitor the buffer zone.

Travel Speed

While travel speed has not, to the best of our knowledge, been explicitly discussed in prior models of buffer zone dynamics, one might anticipate that higher potential travel speeds (*e.g.*, canoes, horses, mechanized transport) might necessitate wider buffer zones. We are unable to systematically determine travel speeds for the examples in our database, so we instead have recorded the most rapid means of transport used in combat in each case. We then code this as a ranked variable reflecting relative speed (foot = 1, canoe = 2, horse = 3, mechanized transportation = 4). We recognize that relative speed may also be influenced by environment and topography; however, it appears to be generally the case that the most rapid available mode of transport for a given environment was typically used.

It should be noted that our data are not a random sample of buffer zones, but rather reflect all the buffer zones we could find in the literature for which we could estimate width and associated population density. We have no reason to suspect that our sample is systematically biased relative to the variables in our sample, *i.e.*, drawn only from particular subsistence modes, or from particular ecologies, political systems, or technological practices. Much of the dataset drawn from the eHRAF sample is based ultimately on the standard ethnographic sample and was therefore designed to produce independent observations of widely spaced cultural contexts. Our historical and modern data may be more biased as they include buffer zones from closely related contexts or multiple buffer zones created by the same state.

Models

Figure 2 displays a directed acyclic graph (DAG) used to model the potential causal structure linking independent variables to buffer zone width, and to design linear models to test the relationship between these variables. DCI and travel speed are modeled as having independent potential effects on buffer width, but the relationship between NPP, density, and subsistence is more complex. NPP is highly likely to impact population density, but mediated by subsistence mode, which determines how much energy can be extracted from a given environment and how many people can therefore hypothetically be supported. NPP may also influence subsistence, however, as some environments may be associated with particular modes of resource extraction (extreme high latitudes with hunting-gathering, or tropical rainforest and horticulture, for instance). That said, some ecologies represented in our data may contain non-industrial subsistence groups practicing a particular mode, but later also industrialized agriculture, so that the relationship may differ depending on how we parse the data. NPP therefore potentially acts as a collider variable between subsistence and buffer width. Consequently, we run four separate linear models examining the potential relationship of independent variables to buffer zone width (see Table 1). Exploratory examination of our variables suggests that at least some of

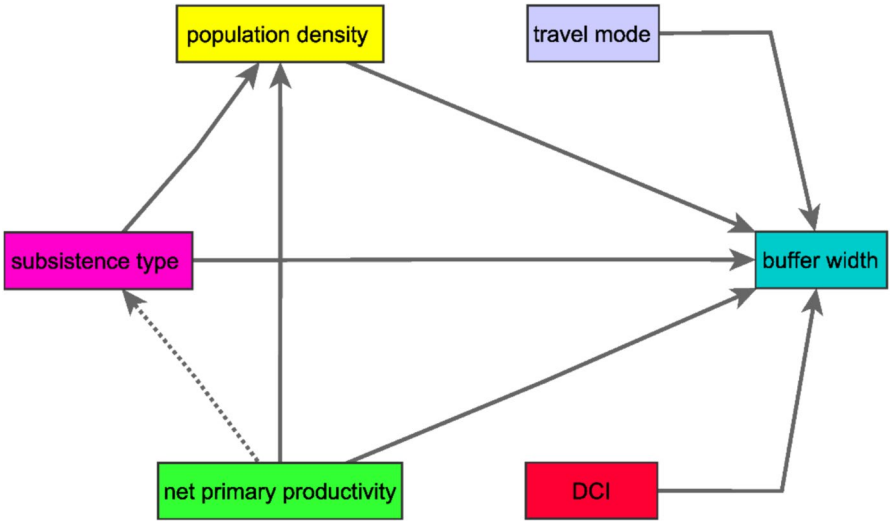

Fig. 2 Directed acyclic graph of possible causal relationships between dependent variables and the independent variable (buffer zone width)

the relationships in our data are only linear when values are logged. Consequently, we have applied a log10 transformation to all of our variables. We also must note that buffer width could in turn impact population density, as the opening of a buffer zone will in principle increase density along its fringes. As such, we cannot treat our models as strict statistical causal inference (Major-Smith, 2023).

Results

All four models produce highly significant adjusted R^2 values indicating potential good model fit. However, it is evident that almost without exception, model fit is driven by the association between population density and buffer zone width. This is true for both the whole dataset and when the analysis is limited to only non-industrial subsistence economies. Considering all cases, no other variable shows any significant association with buffer width. Additionally, likelihood tests comparing the four models indicate that the model including only population density (Model 4) fits the data better than any of the more complex models.

This is largely true also when considering only non-industrial subsistence economies. However, in this case, likelihood tests indicate that Model 3, adding travel speed along with population density, outperforms other models, including Model 4. Model 3 for non-industrialized cases is also the only model for which including an additional variable substantially reduces the association between population density and buffer width. The relationship between population density and buffer zone width is a negative log-log association, so the higher the population density, the narrower the buffer zone (Fig. 3). Buffer zone width drops off at a marginally faster rate with increasing density for industrialized

Table 1 Linear model results. Significant parameter estimates are indicated in bold

All cases	Model 1				Model 2			
Predictors	*ß*	*Estimates*	*CI*	*p*	*ß*	*Estimates*	*CI*	*p*
(Intercept)		**1.19**	**0.65–1.72**	**<0.001**		**1.55**	**1.39–1.70**	**<0.001**
Den	**−0.93**	**−0.59**	**−0.69 to −0.49**	**<0.001**	**−0.95**	**−0.6**	**−0.68 to −0.52**	**<0.001**
NPP	0.08	0.19	−0.10 to 0.48	0.187				
Sub	0.01	0.04	−0.56 to 0.64	0.894				
DCI					0.03	0.62	−1.87 to 3.12	0.612
Travel								
Observations	31				27			
R^2/R^2 adj	**0.928/0.920**				**0.928/0.922**			
Non-industrial only								
Predictors	*ß*	*Estimates*	*CI*	*p*	*ß*	*Estimates*	*CI*	*p*
(Intercept)		**1.29**	**0.22–2.36**	**0.022**		**1.55**	**1.44–1.66**	**<0.001**
Den	**−0.97**	**−0.56**	**−0.70 to −0.42**	**<0.001**	**−0.91**	**−0.52**	**−0.62 to −0.42**	**<0.001**
NPP	0.056	0.14	−0.50 to 0.78	0.648				
Sub	−0.01	−0.03	−0.60 to 0.53	0.907				
DCI					0.14	1.54	−0.43 to 3.51	0.114
Travel								
Observations	19				15			
R^2/R^2 adj	**0.891/0.869**				**0.927/0.915**			

Table 1 (continued)

All cases	Model 3				Model 4			
Predictors	*ß*	*Estimates*	*CI*	*p*	*ß*	*Estimates*	*CI*	*p*
(Intercept)		**1.55**	**1.37–1.74**	**<0.001**		**1.54**	**1.40–1.67**	**<0.001**
Den	**−0.95**	**−0.6**	**−0.68 to −0.53**	**<0.001**	**−0.96**	**−0.61**	**−0.67 to −0.54**	**<0.001**
NPP								
Sub								
DCI								
Travel	−0.02	−0.06	−0.52 to 0.39	0.774				
Observations	31				31			
R^2/R^2 adj	**0.923/0.918**				**0.923/0.920**			
Non-industrial only								
Predictors	*ß*	*Estimates*	*CI*	*p*	*ß*	*Estimates*	*CI*	*p*
(Intercept)		**1.33**	**1.11–1.56**	**<0.001**		**1.52**	**1.41–1.63**	**<0.001**
Den	**−0.77**	**−0.44**	**−0.58 to −0.30**	**<0.001**	**−0.94**	**−0.54**	**−0.64 to −0.44**	**<0.001**
NPP								
Sub								
DCI								
Travel	0.22	0.74	−0.06 to 1.54	0.067				
Observations	19				19			
R^2/R^2 adj	**0.911/0.900**				**0.889/0.883**			

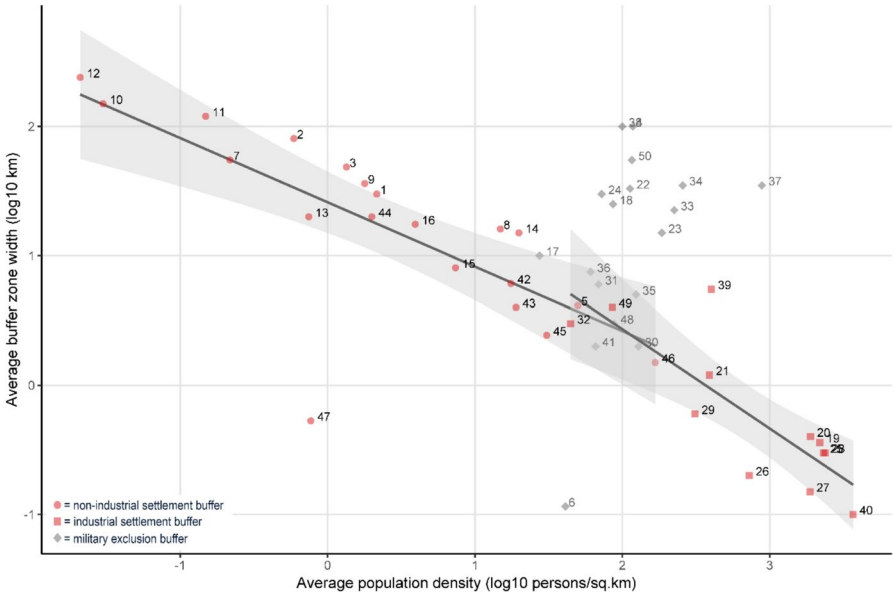

Fig. 3 Average population density (log10 persons/km^2) plotted against average buffer zone width (log10 km), divided by buffer type

cases than for non-industrialized ones, but by and large, the relationship holds across all cases, with the exception of a buffer zone in the Andaman Islands, a case we return to below. Military exclusion zones (which are not included in our formal models) show no association with population density. Neither NPP nor subsistence mode has a discernible independent effect on buffer zone width except to the degree that they impact population density, even when considering only non-industrialized subsistence economies, where we might anticipate the potential value of land to be more important.

DCI also does not show a significant association with buffer width in any of our models, although it has a moderate positive association with buffer width for non-industrialized cases, indicating that the degree of military investment (in terms of the size of war parties or militaries) does not seem to independently significantly impact buffer width beyond what could be explained by population density alone. Travel speed, as noted, does have a moderate positive association with buffer width, but only when considering non-industrialized cases.

Recognizing that our tests cannot offer direct statistical proof of causality, we interpret our results as suggesting that ecological considerations do not play into buffer formation directly. Instead, buffer zone formation appears to be primarily associated with population density, which may be indirectly influenced by resource availability. In non-subsistence economies, travel speed also weakly influences buffer width. What is particularly surprising is how little this relationship varies by subsistence type, or by extension, the diverse ways in which our case studies were politically and socially organized. Regardless of whether combatants are focused on hunting game, small-scale garden horticulture, or practice intensive agriculture

which may entail the ability to extensively transport food (and consequent lack of reliance on the buffer zone itself for resources), population appears to similarly restrict how large a buffer can grow.

Lack of association between DCI and buffer width when added to population density suggests, however, that the relative number of fighters available to monitor the buffer zone actually does not seem to influence buffer width or lead to narrower buffers, as might be anticipated if this were the key factor influencing buffer formation. Consequently, buffer formation appears to be an emergent property of population density itself, although this does not shed light specifically on why or how population serves to constrain how much territory is ceded for defense. It is also not apparent that this need operates in the same way in every instance, given the considerable variability in social structure, land-use, mobility, and so forth that exist in our sample.

Remarkably, this relationship holds even when the motives and strategic objectives of buffer zone formation differ—in most non-industrial contexts, buffer zones function to limit the risk of raiding by making it easier for defenders to become alerted to the presence of raiding parties, increasing the amount of time they have to organize defense (for instance, by gathering fighting men from gardens or other tasks, or signaling to allies), and also by increasing the risks to attackers by making retreat or escape more costly in the event that they are discovered. Buffer zones may also serve this function in state contexts, but in our reading of the literature, this is largely the case when states are concerned with non-state actors or militias that fight like "tribal" war groups. Otherwise, buffer zones seem to be largely used to monitor or control the movement of people across state boundaries, or to limit the likelihood of accidental skirmishes reigniting larger-scale fighting (the Korean DMZ, for instance). The later motivation is the primary objective of military exclusion zones, although Beehner and Meibauer (2016) suggest that they are largely ineffective and that both military and settlement buffers have consequently been little employed in modern warfare. This suggests that the dynamics of relocating population itself plays a critical role in determining buffer width.

The formation of buffer zones has been observed and described in some ethnographic contexts. For instance, the ethnographer Karl Heider observed the formation of buffer zones between groups of allied Dugum Dani villages in the highlands of western New Guinea (documented in Robert Gardener's 1963 ethnographic film *Dead Birds*). Buffer zones among the Dani shifted as alliances were made and remade. When conflict across a particular social division ceased, abandoned land was resettled, with garden plots and hamlets located in newly contested territory abandoned. Watch towers were built along the edges of garden plots to watch for the approach of raiding parties. There is some indication that buffer width was sometimes deliberately determined based on past experience; however, in most non-state contexts, buffer zones appear to arise organically as people progressively move out of areas that become too dangerous to occupy (Gardner & Heider, 1969; Heider, 1997; Roscoe, 2011).

In state contexts, settlement buffers are more commonly created by formal agreement (*e.g.*, the Korean DMZ), or may be enforced unilaterally by one side of an ongoing conflict to prevent incursions from the other or to regulate movement of

people (*e.g.*, West Bank Exclusion Barrier, Berlin Wall, Inner-German Barrier). In these cases, settlements are cleared and relocated by the relevant state actors, often by force. However, buffer zones may also be created by agreement or custom in non-industrial contexts. For instance, among the Luo of Kenya and Tanzania, customary patterns of landownership included "thim," areas of unoccupied grazing land up to ten miles wide between settlements that were maintained as part of customary land ownership, but also were the location of fighting between maximal lineages (Evans-Pritchard, 1949; Wilson, 1961). After colonial pacification, these areas were in many cases converted into agricultural land (Birman, 2024).

A variety of factors might influence why buffer zones are so constrained by population density. This includes issues of resource stress, *i.e.*, the point at which ceding more land becomes incompatible with producing enough food, but also social factors like the ability to move further back without encroaching into the customary or effective territories of allies or other enemies. Relocating likely also entails expending social capital in the form of needing to move into territory used by other communities, for instance acquiring garden land or rights to hunt in particular areas. In many contexts, buffer growth produces settlement aggregation (DeBoer, 1981; Rowlands, 1972), which might lead to issues of scalar stress as communities grow too large to remain socially and economically functional and stable (Ferguson, 1992). Implicit in scalar stress are also political dynamics, which may in part account for why both state and non-state buffers follow a similar pattern of drop-off, *i.e.*, the need to negotiate relocation, even if the specifics of such negotiations will vary by context. Any or all of these factors might contribute to buffer zones stabilizing at a width consistent with population density.

The only case in our ethnographic and historic dataset that diverges significantly from the modeled relationship between buffer width and density may serve as the exception that proves the rule, demonstrating how resource circumscription may limit buffer growth. On South Andaman Island, population densities are estimated as ~ 1 person/km^2, for which we would predict a buffer width of ~ 50 km. However, ethnographically documented buffers between the Bea and Jarawa during the earlier twentieth century are only on the order of a third of a kilometer (Kelly, 2000, 2005) South Andaman Island has a maximal north-south width of ~ 40 km and an E-W width of ~ 20 km (#47 on Fig. 3). Clearly, the inhabitants of the island would be unable to open a buffer zone even a fraction of the predicted width given the immediate resource issues this would entail. These buffers may therefore represent the maximal distance that could reasonably be opened for security given the immediate resource issues that larger buffers would create.

Modeling and Identifying Ancient Buffer Zones

While buffer zones are unlikely to directly inform on the underlying causes of collective violence in the past (allowing for the possibility that their formation could generate subsistence stress), ancient buffer zones may still provide evidence for the existence of conflict in the past, as well as particular dynamics such as the nature of combatants and duration of hostilities. That said, identifying

buffer zones in the archaeological record is by no means a straightforward task. As a general issue, identifying a buffer zone in the archaeological record relies intrinsically on negative evidence. A variety of factors could account for the lack of settlement evidence, including lack of preservation, biases in archaeological knowledge, unsuitability of territory (which may vary depending on how people use land), or cultural avoidance of particular areas (due to association with ancestral ghosts, or other specific beliefs). As such, careful consideration must be made of other possible reasons for a gap in the settlement record, combined with other evidence for conflict. Watrall (1968) for instance, was able to demonstrate the paucity of eighteenth-nineteenth century material culture in the area of the buffer zone described by Hickerson, albeit with historical knowledge of its location and duration.

DeBoer (1981) provides the only general framework that we are aware of to systematically outline how buffer zones might be located and delimited using archaeological data, using linear plots of site size, and arguing that larger aggregated settlements should occur at the edges of buffers. That said, it is not always clear that settlements should extend to the limits of buffer areas—among the Dugum Dani, for instance, villages were set back from buffer fringes, which were lined by garden plots and guarded by watch towers (Heider, 1997). DeBoer notes other basic issues of identifying buffers, including archaeological visibility and geomorphological processes that could bias understanding of settlement distribution.

DeBoer also highlights the role of buffer zone stability in archaeological visibility, or viewed another way, how quickly buffer zones shift in location. Buffer zones, like other forms of defense (Rowlands 1972), manifest the military organization and integration of combatants. Particularly in non-hierarchically politically integrated contexts, the details of alliance (Arkush *et al.*, 2025), confederacy (Birch *et al.*, 2025), and kinship may play important roles in understanding where buffer zones form and how durable they are. For instance, the customary "thim" of the Luo existed within a context of nested patrilineages in which kinship, community, and military alliance boundaries were relatively stable and more or less overlapped. The Dugum Dani, meanwhile, lived in clustered hamlets that were less rigidly structured by kinship and between which there were more fluid relationships, and military and alliance structures were less intrinsically tied to location as a result (Heider, 1997). Of course, kinship is a more diverse and flexible system than such a simple dichotomy would allow for, and can also shift in response to violence. In any case, it is likely that more stable and culturally entrenched buffers are likely to have far higher archaeological visibility, and so the ability to even recognize a buffer zone in the archaeological record may reflect underlying dynamics of social and military relationships that produce more persistent and durable buffer areas. As DeBoer (1981) notes, buffer zones are also less likely to persist in the event that one side has either an intrinsic military advantage (*e.g.*, far higher population) or gains one (the acquisition of new technology, for instance), in which case any latent buffer zone may close as quickly as the disadvantaged side tries to open it. We hypothesize that buffer zones such as those we have identified represent situations in which a relative military stalemate has been reached, at least for a short period of time, or where

shifting alliances mean that no side is able to sustain a long-term demographic and military advantage.

A related issue is the level of archaeological spatial and chronological resolution, and how these relate to the chronological and geographical scale of expected buffer areas. For instance, archaeologists working in the US Southwest are able to use highly detailed ceramic seriations combined with dendrochronology to identify areas in which settlement is temporarily absent in the context of other indicators of violence (Wallace & Doelle, 2001; Wilcox & Robertson, 2001). Marcus & Flannery (1996: 124, Figure 128) argue for the opening of a buffer zone between competing minor polities during the Middle Preclassic Rosario Phase (~700 to 500 BC) in the northern Valley of Oaxaca, Mexico. This buffer zone was eventually occupied after a military victory that resulted in the consolidation of the Zapotec State and the emergence of its primary urban center at Monte Albán in the center of the former buffer zone. In this case, they rely on decades of intensive field survey coupled with excellent archaeological visibility and a detailed ceramic sequence to identify settlement shifts that may have occurred over a few decades. In regions where chronology is less well understood, less survey has been conducted, or the environment is less favorable to site recognition, it may be more challenging to persuasively identify a potential buffer zone—it is also worth noting that both the American Southwest and Oaxaca are relatively arid environments with relatively good ground visibility.

While we doubt that criteria exist that can be universally applied for identifying and mapping ancient buffer zones, we would suggest a number of "rules of thumb" that should be considered. Identifying a buffer zone is not so much a matter of finding evidence as testing a hypothesis that can then inform further on the dynamics of conflict if a plausible argument can be built. Other possible explanations for a settlement gap should be evaluated, including biases in survey and understanding of settlement distribution, taphonomic or geomorphological factors that may obscure settlement, as well as possible social or economic reasons why settlement may be absent from a particular area. This also entails considering whether the degree of chronological resolution is sufficient to understand relatively shorter-term shifts in buffer zones (*e.g.*, a gap that exists for hundreds of years is less plausible than one that opens and persists for decades or a century). While our data do not specifically bear on this issue, we have no reason to suspect that buffer zones are persistent and ubiquitous features of non-industrial land-use based on our results, but rather the specific outcome of ongoing conflict. As such, they should not be expected (at least of the type we are discussing) unless other plausible signs of conflict (burial trauma, site enclosures, *etc.*…) are also present. It may be necessary to consider the utility of buffer zones more broadly within the context of expected forms of violence given knowledge of social organization and other relevant factors, including the organization of fighting (*i.e.*, informal raiding parties vs. standing armies).

Finally, population reconstructions, to the extent that they are available (and allowing for the challenges of reconstructing ancient population—Drennan *et al.*, 2015) can be used to understand the plausible size of buffer zones in any particular case study. This may be as coarse-grained as suggesting that a buffer zone of a hundred kilometers in width is unlikely to occur in the context of relatively intensive horticulture, for instance. However, the modeled relationship between density

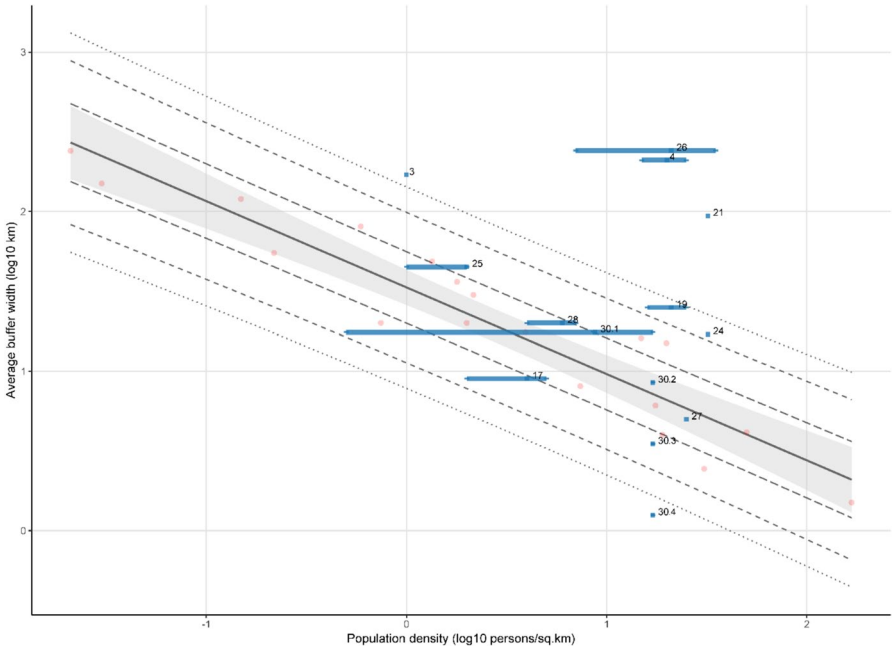

Fig. 4 Average population density (log10) vs. average buffer width (log10). 1-, 2-, and 3-sigma prediction bands derived from ethnographic and recent historical cases are indicated by dashed lines. Published estimated population density ranges for archaeological and historical buffer zones are indicated by blue bars

and buffer width can also be used to evaluate possible buffer areas if population estimates are available. Figure 4 shows a number of archaeological and proto-historic cases plotted against a linear model comparing logged population density and logged buffer width including only the non-industrial cases in our database, with the Andaman Islands removed. We do not claim to predict an exact buffer width given the challenges of reconstructing ancient populations and exactly measuring settlement gaps, but instead use this as a heuristic for determining how plausible a certain possible buffer zone might be given broader archaeological understanding of socio-economic organization and approximate population size.

To illustrate this approach, we first present an example where the presence of buffer zones was noted historically, but in a context for which most knowledge comes from the archaeological record. The Coosa chiefdom of the Georgia/Alabama/Tennessee border area (Fig. 5) was historically documented by members of the DeSoto expedition (AD 1539–1543) as well as two subsequent Spanish expeditions to the area later in the sixteenth century (Hudson, 1997; Smith, 2000). This paramount chiefdom was described as conducting raids against neighbors to extract tribute, while village sites were described as enclosed by palisades and surrounded by buffer areas. Archaeologically, the Coosa chiefdom has been linked to the Carters site cluster in northwestern Georgia, while other locations described in Spanish accounts are likely associated with a number of other contemporary site clusters

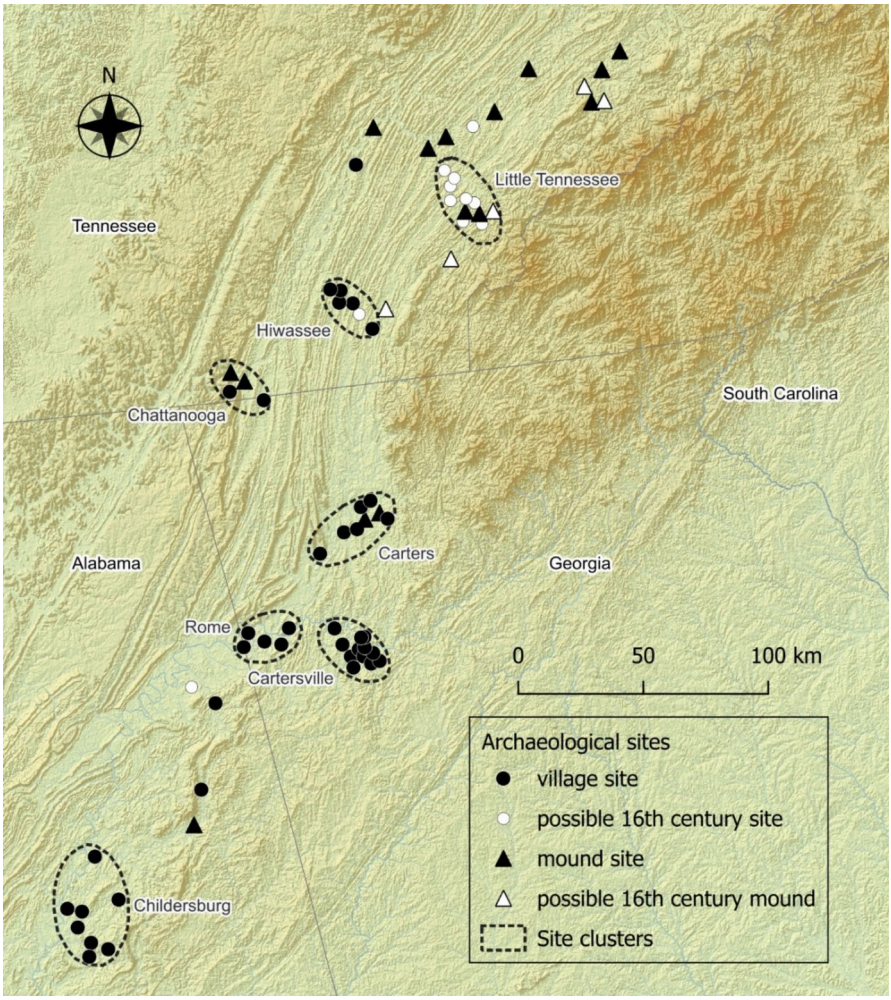

Fig. 5 Sixteenth century AD site distribution in the Tennessee River Valley and surrounds. Redrawn after Smith, 2000: Figure 18

located along the Tennessee River Valley and on its fringes. These clusters are separated by gaps of ~ 10 to 30 km that may align with these historically described buffers, while at least some village sites that were archaeologically investigated are surrounded by palisades and ditches. These unoccupied areas do not appear to differ ecologically from occupied areas (Hally *et al.*, 1989; Smith, 2000).

Our ethnographic model would predict a population density of roughly 2.5 persons per square kilometer based on an average of ~ 20 km for buffer areas. Hally and colleagues (1989) provide estimates for houses in each site cluster based on site area and estimating 5.9 and 11 persons per house on average to arrive at population estimates. We have converted these density estimates by dividing by the total

area within each delimited site cluster (assuming utilized fields between villages, as historically described) to arrive at a density range of between 3 and 7 persons per square kilometer (#28 on Fig. 4). While there are of course numerous assumptions built into both the density estimates and our estimation of buffer width (*i.e.*, that buffers extend to the limits of settlement), this estimate is at least roughly consistent with what we might expect given both the modeled population densities and expected buffer widths. Sattler (2009) also describes buffer zones of between 1.6 and 8 km between segments of the Creek Confederacy during the eighteenth century in the same region. Populations in the American Southeast are believed to have increased by the eighteenth century after a late sixteenth century decline (Kowalewski & Hatch 1991), cf. Holland-Lulewicz *et al.*, 2020), and Sattler estimates a density of ~ 25 persons/square kilometer from historical data, consistent with the ~ 24 persons/square kilometer our ethnographic model would predict (#27 on Fig. 4). While not proving that settlement gaps during the sixteenth century were buffer zones, their width and the apparent dynamics of settlement gaps in the region are consistent with our model.

Our second case study examines potential buffer zones between early Neolithic *Linearbandkeramik* (LBK, ~ 5400 to 4900 calBC) villages in eastern Belgium (Fig. 6). While Keeley (1997, see also Golitko & Keeley, 2007) has argued that much early Neolithic violence, including in eastern Belgium, occurred between newly settled LBK farming communities and longer resident foragers, most recent scholarship suggests that conflict likely primarily occurred between LBK communities themselves (Fibiger *et al.*, 2023). Archaeological evidence suggests that conflict took the form of endemic raiding punctuated by less frequent larger scale attacks that in some cases may have wiped out significant portions of village populations (Duering & Wahl, 2014; Golitko & Keeley, 2007; Meyer *et al.*, 2018). Some of these villages were enclosed with ditches and palisades, although the purpose of LBK enclosures remains debated (Golitko & Keeley, 2007). Given the indisputable evidence for intense inter-community violence, buffer zones might be an anticipated form of defense. LBK sites tend to occur in large aggregated clusters termed *Siedlungskammern* separated by unoccupied land. While LBK villages are often modeled as politically acephalous, this does not rule out the possibility that they banded together into confederations or alliances at times (Birch *et al.*, 2025). *Siedlungskammern* have been argued by some to represent loose social or political coalitions, including the possibility that they represent military alliance groups (Keeley & Cahen, 1989).

In eastern Belgium, LBK sites were first established around 5200 calBC, probably by migrants originating at LBK villages further east in the Dutch Limburg or Germany. On present evidence, these early settlements consisted of one or two characteristic longhouses. However, after a relatively short time, some of these hamlets developed into larger communities surrounded by substantial ditch and palisade systems (Golitko, 2015). Remote sensing surveys (Quick, 2010) have failed to document enclosures beyond those shown on Fig. 6. Sites in eastern Belgium also group into two *Siedlungskammern*, with a third cluster of sites further to the east in the Dutch Limburg. While this latter cluster is separated from Belgian LBK sites by the channel of the Meuse (Maas) River, much of the rest of the area is blanketed in

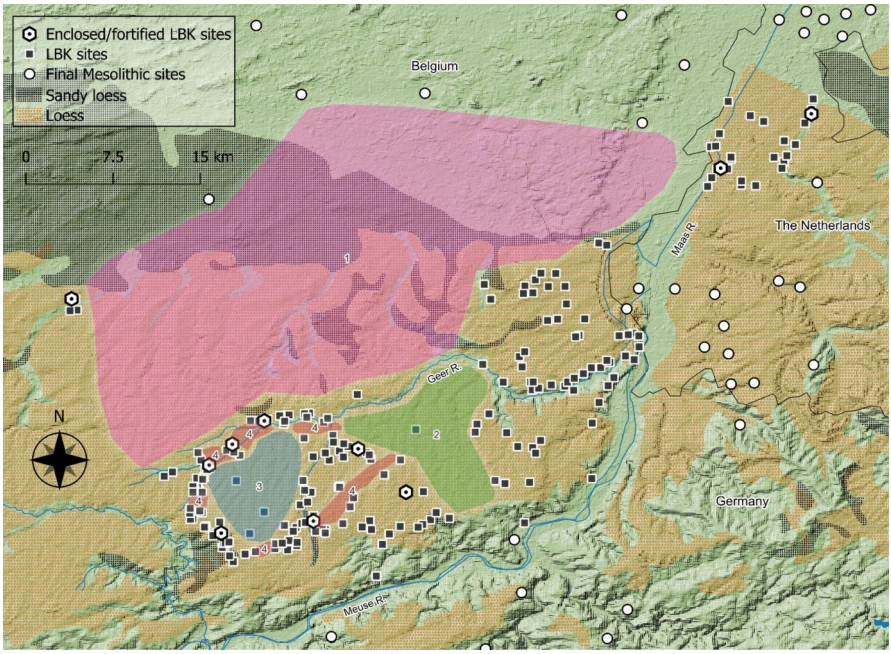

Fig. 6 Map of the earliest Neolithic (LBK) and final Mesolithic sites in eastern Belgium and the Dutch Limburg with the distribution of loessic sediments shown. Enclosed/fortified LBK villages are indicated. Redrawn with modifications after Keeley 1997: Figure 11.3

loessic soils that were preferred for settlement during the early Neolithic, providing no obvious ecological reason for other regional gaps in settlement.

However, there remain uncertainties in both the chronological development of settlement in the region and how conflict is conceptualized that make it less obvious at what scale one would anticipate buffer areas. It is, for instance, unclear whether small hamlets were abandoned as larger fortified settlements grew, or if these fortifications served as refuges for clusters of surrounding sites (Quick, 2010), as most excavation has focused on larger fortified settlements. The scale of social and possibly military cohesion has also been modeled at a variety of scales. As noted, Keeley argued that conflict largely occurred between hunter-gatherers to the north, with the *Siedlungskammer* as a basic unit of social and military alliance, but conflict could also have been between *Siedlungskammern*. One of us has argued for shifting internal alliances within *Siedlungskammern* based on differences in resource procurement and trade patterns (Golitko, 2015). While Quick (2010) follows Keeley in arguing for conflict between LBK farmers and foragers to the north, his settlement model could also, in principle, allow for these clusters of aggregated sites surrounding fortifications to represent alliance groups or other opposed combatants.

This allows for several possible buffer zones that could have been present in eastern Belgium at the end of the sixth millennium BC. The first of these is the ~ 15 km-wide buffer proposed by Keeley (1 on Fig. 6), but the gap between *Siedlungskammern* (2 on Fig. 6) could alternatively be modeled as a buffer zone, as could a gap

between LBK sites located in the Upper Geer and Yerne drainages respectively (3 on Fig. 6), or gaps between fortification focused site clusters (4 on Fig. 6). The distribution of enclosed sites could in principle be viewed as consistent with any of these models. Various population densities have been estimated for the central European Neolithic (LBK), ranging between ~ 1.5 persons/square kilometer up to 17 persons/square kilometer depending on whether reconstructions are based on total settled area or restricted to loessic soils and/or settlement areas only (Lüning, 1988, 1998; Schmidt *et al*. 2021; Zimmermann, 1996; Zimmermann *et al.*, 2009). These values are based on house population estimates for the Aldenhovener Platte region of Germany, where open-cast coal mining allowed the total mapping of LBK settlement. Mesolithic population densities in temperate Europe are estimated to be well under one person per square kilometer, although this density may have increased by the early Atlantic period when agriculture first appeared and is an average value for all of inland Europe (Gronenborn, 1999; Jochim, 1976). Of the potential buffer zones, that between *Siedlungskammern* is most consistent with estimated population density (see #30.2 on Fig. 4), although it should be noted that the lack of any larger water courses in the region of the potential buffer zone may account for the lack of settlement (Quick, 2010). Given uncertainties in estimates of population density, any of the proposed buffers could potentially be consistent with the modeled relationship (see #30.1, 30.3, and 30.4 on Fig. 4), and it of course remains possible that none of these areas are buffer zones. In this case, further understanding of more localized population densities, as well as site contemporaneity, may be needed to further assess the likelihood that these gaps may have served as buffers.

There are a number of proposed buffers in our survey of the archaeological literature that are less consistent with our modeled relationship between buffer width and population density. This could indicate that buffer widths are misestimated, population density estimates are substantially off, or that the proposed buffers are not in fact settlement buffers opened in response to conflict (or a combination of factors). Notably, this includes a prehistoric buffer zone proposed by DeBoer for the Amazon Basin between the Cumancaya and Caimito archaeological traditions of the upper Ucayali during the thirteenth century AD (#4 on Fig. 4). An ethnohistorically documented buffer between the Cocima and Conibo in the region (sixteenth century, #3 on Fig. 4) is consistent with DeBoer's estimates of ~ 1 person/square kilometer derived from contemporary reports. This density is also consistent with more recent reconstructions of population density in the region (Denevan, 2003, 2014). DeBoer also documents how these historic buffer zones shifted and fluctuated during the 16th and 17th centuries as alliances and population shifted in response to endemic conflict. These buffer zones also are far wider in the far western Amazon Basin than further east, consistent with modern population density distribution.

However, DeBoer's estimate of 15–25 persons/square prior to European arrival is an order of magnitude higher than the 210 km buffer he reconstructs for the thirteenth century would predict, and of similar size to those that existed in the region during the historic period. This higher population density is more consistent with recent archaeological findings which suggest far higher Amazonian populations than once reconstructed. That said, the *terra preta* (anthropogenic "dark earth") believed to indicate particularly intensive longer-term settlement in Amazonia are relatively

less frequent in the Upper Ucayali than further to the east (McMichael *et al.*, 2014; Palace *et al.*, 2017), so it is plausible that DeBoer's population estimate in this case was too high. This does not rule out the possibility that the settlement gap he identifies is not a buffer zone, or even an artifact of limited archaeological coverage or other factors. It is not our intention here to argue for or against the accuracy of this particular archaeological reconstruction, or enter the debate about prehistoric population sizes in the Amazon Basin. Rather, we use this case to further show how our proposed approach can be used to evaluate the plausibility of buffer zones in the archaeological record using a set of expectations that appear to be independent of the particulars of region or period, and can therefore be applied to almost any archaeological context in which there is evidence for collective violence.

Conclusion

While buffer zones (areas unoccupied owing to conflict) have been modeled as intrinsic components of population-resource dynamics, our analysis suggests that their ecological value has little to do with the dynamics of their formation. Instead, they are highly constrained by population density, and in non-industrial contexts, respond to travel speed. This suggests that they primarily form as a result of security concerns, with any ecological benefits or drawbacks secondary to this primary motivation. This relationship holds across a wide range of ecologies, political types, military organizations, and cultural practices. We suggest that they may be particularly useful in the context of the opportunistic raiding that characterizes much non-industrialized collective violence, although more hierarchically organized actors may still utilize buffer zones to control population movement or as a defensive measure against less hierarchically organized combatants. Buffer zones can be anticipated as an outcome of ancient conflict, and if identified, may shed light on the size or constitution of alliances, coalitions, and other forms of confederacy that are otherwise difficult to identify archaeologically (Arkush *et al.*, 2025; Birch *et al.*, 2025). However, recognizing buffer zones in the archaeological record is challenging, as they intrinsically represent a form of negative evidence. To argue for a buffer zone, archaeologists must demonstrate a plausible context in which they may have occurred, eliminating other potential reasons for a real or apparent gap in an archaeological settlement record. Allowing for difficulties in estimating ancient population density, the estimated size of ancient populations may also help to understand whether a settlement gap could plausibly have been a buffer zone.

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Author Contribution MG wrote the main manuscript text. MG and DMG conceptualized the research design. MG and DMG collected data. MG produced all figures. MG and DMG reviewed the manscript.

Data Availability All data are provided in the supplemental file.

Declarations

Competing Interests The authors declare no competing interests.

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